CRYPTOCURRENCY PRICE PREDICTION

1. INTRODUCTION

Cryptocurrency markets are among the most volatile and dynamic financial ecosystems in the world today. Unlike traditional financial instruments, cryptocurrencies operate in a decentralized manner, free from central bank control, and are influenced by a complex interplay of factors including market demand, investor sentiment, global economic conditions, regulatory developments, technological innovations, and trading volume. The price of Bitcoin, the most prominent cryptocurrency, has experienced dramatic fluctuations since its inception, rising from mere cents to tens of thousands of dollars, making accurate price prediction an exceptionally challenging yet critically important task for investors, traders, and financial analysts alike.

The analysis is performed on Bitcoin (BTC-USD) historical price data from January 1, 2018 to the present. By evaluating and comparing the forecasting performance of these models using standard evaluation metrics, this project aims to identify effective approaches for short-term cryptocurrency price prediction.

1.1 Objective

- To study historical Bitcoin price data.
- To analyze Bitcoin market trends and price movements.
- To predict Bitcoin prices using ARIMA, SARIMA, Prophet, and LSTM models.
- To compare the performance of different prediction models.
- To forecast future Bitcoin prices for short-term analysis.
- To understand the effectiveness of machine learning and time series models in cryptocurrency prediction.

2. DATASET DESCRIPTION

The dataset used in this project was collected through the Yahoo Finance API (yfinance), which offers reliable and well-structured access to historical cryptocurrency price data over a specified time period. The data encompasses Bitcoin (BTC-USD) daily trading information from January 1, 2018 to the present date (13 May, 2026), covering more than eight years of market activity. This timeframe includes several significant market events, including the crypto winter of 2018, the COVID-19 crash and recovery of 2020, the bull run of 2021, and subsequent market corrections, providing a rich dataset for model training and evaluation.

2.1 Features

The dataset includes the following five core features that characterize daily Bitcoin trading activity:

Table 1. Dataset Feature Descriptions

Feature	Description
Open Price	The price at which Bitcoin started trading for the day
High Price	The highest price reached during the trading day

Low Price	The lowest price recorded during the trading day
Close Price	The final price at the end of the trading day
Volume	The total amount of Bitcoin traded during the day

For this project, the main focus is on the closing price, as it reflects the final market consensus for the day and is the most commonly used metric for price prediction tasks. The closing price serves as the primary target variable for all four forecasting models, while additional engineered features such as moving averages and daily returns are used to enhance the exploratory analysis and provide deeper insight into market behavior.

3. DATA PREPROCESSING

3.1 Data Quality Assessment

Before applying any modeling techniques, a thorough data quality assessment was conducted to ensure the reliability of the dataset. The analysis confirmed that the dataset contains zero null values and zero duplicate entries across all features. This is largely attributable to the quality of the Yahoo Finance data source, which maintains consistent daily records for actively traded cryptocurrencies. The absence of missing values eliminates the need for imputation strategies, and the lack of duplicates ensures that no data point receives undue weight during model training.

Quality Check	Result
Null Values	0 (all features)
Duplicate Rows	0
Date Range	Jan 1, 2018 – May 13, 2026
Frequency	Daily

3.2 Feature Engineering

To enhance the analytical depth and provide additional perspectives on market behavior, several derived features were engineered from the raw closing price data. These features help capture both short-term and long-term market trends, as well as the volatility characteristics of Bitcoin prices:

50-Day Moving Average (MA): Calculates the average closing price over the past 50 trading days, providing a smoothed representation of medium-term price trends. This indicator helps filter out daily noise and reveals the underlying directional momentum of the market. When the current price is above the 50-day MA, it typically signals bullish conditions, while trading below suggests bearish sentiment.

200-Day Moving Average (MA): Calculates the average closing price over the past 200 trading days, serving as a key indicator of long-term market trends. The 200-day MA is widely followed by institutional investors and is considered one of the most important technical indicators in financial analysis. The relationship between the 50-day and 200-day MAs forms the basis for identifying Golden Cross and Death Cross signals, which are among the most closely watched technical patterns.

Daily Return: Measures the percentage change in closing price from one day to the next, calculated using the formula: Daily Return = (Close Price today Close Price yesterday) / Close Price yesterday.

This metric is essential for understanding price volatility and risk, as higher absolute daily returns indicate greater market uncertainty and potential trading opportunities.

4. EXPLORATORY DATA ANALYSIS (EDA)

4.1 Bitcoin Price Trends Over Time

The historical Bitcoin price chart reveals the dramatic price movements that have characterized the cryptocurrency market over the analysis period. From its early 2018 levels following the peak of the 2017 bull run, Bitcoin experienced a prolonged bear market throughout 2018 and into early 2019. The market then showed signs of recovery before the COVID-19 pandemic triggered a sharp but brief sell-off in March 2020. This was followed by an extraordinary bull run that saw Bitcoin reach new all-time highs in 2021, followed by another significant correction and subsequent recovery cycles. The overall trajectory demonstrates both the tremendous growth potential and the extreme volatility inherent in cryptocurrency markets.

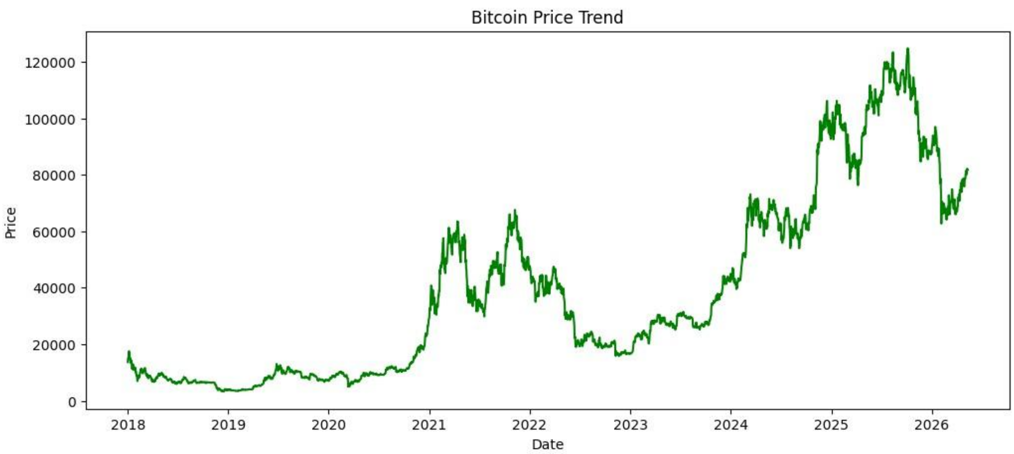

Figure 1. Bitcoin Price Trend (2018 - 2026)

4.2 Stationarity Analysis

Stationarity is a fundamental requirement for many time series models, particularly ARIMA and SARIMA. A stationary series has constant mean, variance, and autocorrelation structure over time, which ensures that the statistical properties used for modeling remain consistent across the entire dataset. The Augmented Dickey-Fuller (ADF) test was applied to assess the stationarity of the Bitcoin closing price series. The test produced an ADF Statistic of -1.006 with a p-value of 0.751, which is well above the significance threshold of 0.05. This result confirms that the raw Bitcoin price series is not stationary, meaning it exhibits time-dependent trends and must be differenced before applying statistical time series models.

Test Metric	Value	Interpretation
ADF Statistic	-1.006	Not sufficiently negative
p-value	0.751	Far above 0.05 threshold
Conclusion	Non-stationary	Differencing required for ARIMA/SARIMA

Table 3. Augmented Dickey-Fuller Test Results

4.3 Moving Average Analysis

The overlay of the 50-day and 200-day moving averages on the Bitcoin price chart reveals important technical patterns that have historically served as reliable indicators of market direction. When the 50-day moving average crosses below the 200-day moving average, the pattern is known as a Death Cross, which typically signals a potential bearish trend and has often preceded significant price declines. Conversely, when the 50-day moving average crosses above the 200-day moving average, the resulting Golden Cross pattern signals a potential bullish trend and has historically been associated with sustained price increases. These crossover events are closely monitored by traders and investors as they provide insight into shifting market momentum and can inform strategic entry and exit points for positions.

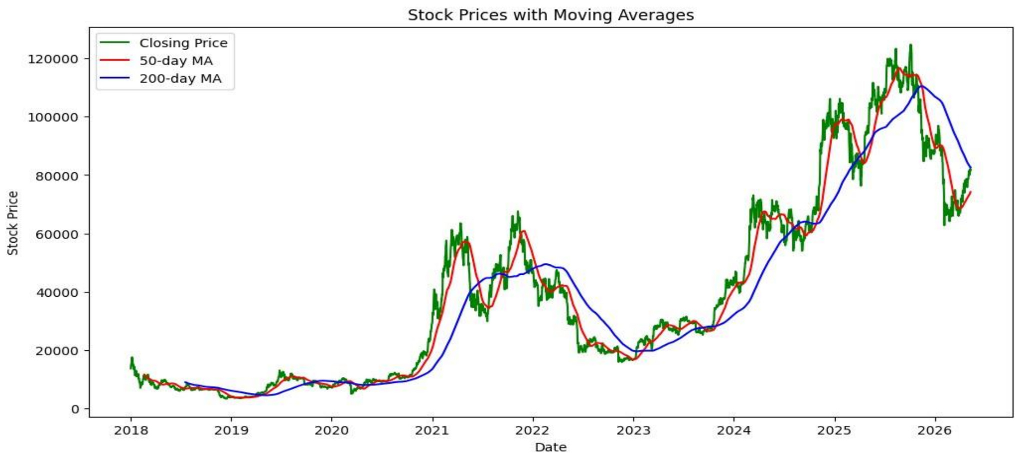

Figure 2. Bitcoin Price with 50-Day and 200-Day Moving Averages

4.4 Volatility Analysis – Daily Returns

The daily returns plot provides a visual representation of Bitcoin price volatility over time. The graph shows the percentage change in price from one day to the next, with values fluctuating around the zero line. Several key observations emerge from this analysis: First, the returns exhibit significant clustering, where periods of high volatility tend to be followed by additional high-volatility periods, a phenomenon known as volatility clustering that is common in financial markets. Second, sharp spikes both above and below the zero line indicate periods of extreme market movement, often triggered by major news events, regulatory announcements, or large institutional trades. Third, the overall magnitude of daily returns underscores the inherent risk in cryptocurrency trading, with daily price swings of 5-10% occurring with notable frequency during turbulent market conditions.

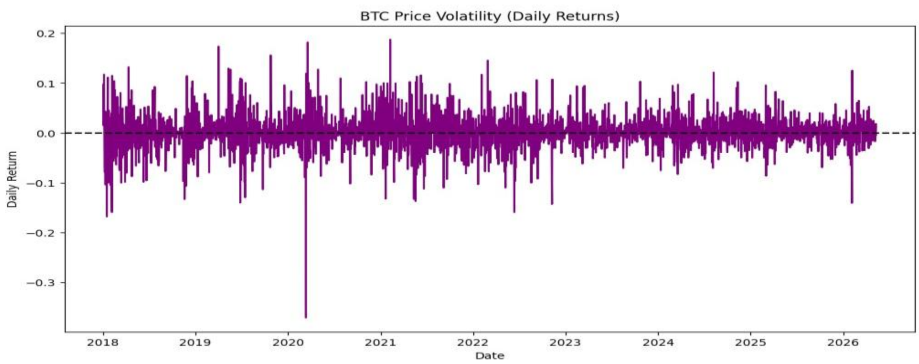

Figure 3. Bitcoin Price Volatility (Daily Returns)

4.5 Data Distribution Analysis

Box plot analysis of the dataset features reveals the distributional characteristics and presence of outliers across all variables. The price-related features (Open, High, Low, Close) show similar distributions with significant upward skew, reflecting the overall upward trend in Bitcoin prices over the analysis period. The Volume feature displays numerous outliers on the upper end, indicating occasional days with exceptionally high trading activity, typically coinciding with major market events. The 50-day and 200-day moving averages show narrower distributions as expected, given their smoothing properties. The Daily Return feature exhibits a roughly symmetric distribution centered near zero, consistent with the efficient market hypothesis, but with heavy tails that reflect extreme market movements beyond what would be expected under a normal distribution.

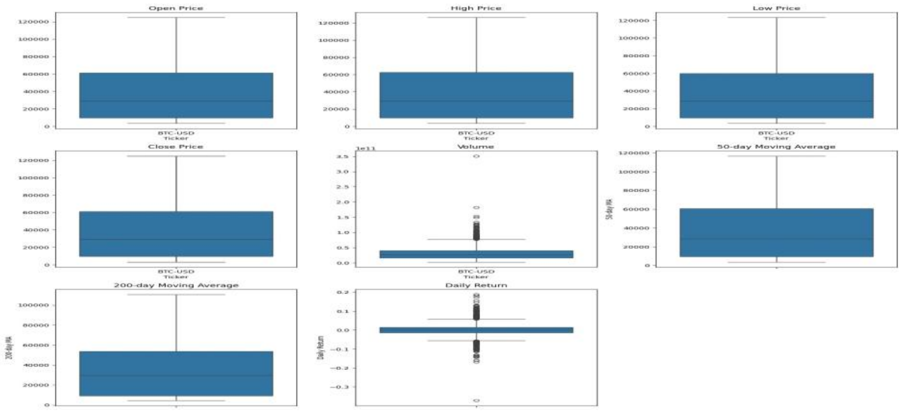

Figure 4. Box Plot Analysis of Dataset Features

5. ARIMA MODEL

ARIMA (Autoregressive Integrated Moving Average) is a statistical time series model widely used for forecasting future values based on past observations. The model combines three key components: Autoregression (AR), which uses the relationship between an observation and a specified number of lagged observations; Integration (I), which uses differencing of raw observations to make the time series stationary; and Moving Average (MA), which uses the dependency between an observation and residual errors from a moving average model applied to lagged observations. The ARIMA model is particularly effective for capturing linear patterns in time series data and is commonly used for short-term forecasting tasks.

5.1 Model Configuration

The optimal ARIMA parameters were determined using Auto ARIMA, an automated parameter selection method that evaluates multiple combinations of (p, d, q) values and selects the best model based on the Akaike Information Criterion (AIC). The search identified ARIMA (1,1,0) as the optimal configuration with an AIC value of 52,609.789. This configuration indicates that the model uses one autoregressive term, one order of differencing to achieve stationarity, and no moving average component. The data was split into 80% for training and 20% for testing, with the model being updated step-by-step during testing through a rolling forecast approach where each new actual value is incorporated into the training set before making the next prediction.

Parameter	Value	Description
p (AR order)	1	One autoregressive term
d (Differencing)	1	First-order differencing
q (MA order)	0	No moving average component
AIC	52,609.79	Best among tested configurations
Train/Test Split	80/20	Rolling forecast approach

Table 4. ARIMA Model Configuration

5.2 Prediction Results

The ARIMA model demonstrates strong performance in predicting Bitcoin prices on the test dataset. The visual comparison between actual and predicted prices shows that the model closely tracks the actual price movements, with only minor deviations during periods of rapid price change. The rolling forecast approach, where the model is retrained with each new observation, contributes significantly to this accuracy by allowing the model to adapt to evolving market conditions.

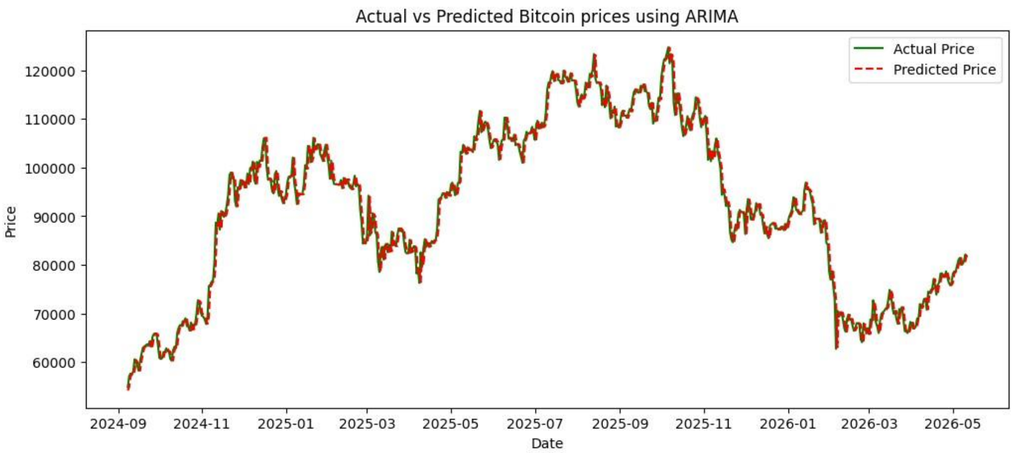

Figure 5. Actual vs Predicted Bitcoin Prices using ARIMA

5.3 Performance Metrics

The quantitative evaluation of the ARIMA model confirms its strong predictive capability across multiple error metrics. The model achieved a Mean Absolute Percentage Error (MAPE) of just 1.68%, indicating that predictions deviate from actual values by less than 2% on average. The R-squared score of 0.985 demonstrates that the model explains approximately 98.5% of the variance in Bitcoin prices, reflecting an excellent fit to the data. These results establish the ARIMA model as a reliable baseline for short-term cryptocurrency price forecasting.

Metric	Value	Interpretation
MSE	4,397,411	Low squared error relative to price scale
RMSE	2,097.00	Approx. 2.3% of average BTC price
MAE	1,506.52	Average absolute deviation in USD
MAPE	1.68%	Excellent prediction accuracy
R-squared	0.985	98.5% of variance explained

Table 5. ARIMA Model Performance Metrics

5.4 30-Day Forecast

Two methods were employed for generating 30-day Bitcoin price forecasts: the direct multi-step forecast and the recursive step-by-step forecast. The direct method projects all 30 future values at once using the trained ARIMA model, while the recursive method predicts one day at a time, appending each prediction to the dataset before generating the next. Both methods predict a relatively stable price trend in the near future, with only slight variations. The direct ARIMA forecast predicts slightly higher prices compared to the recursive forecast, but both follow a similar pattern, indicating consistency between the two approaches. The direct method is more efficient and suitable for practical use, while the recursive method may introduce slight deviations due to error accumulation over time.

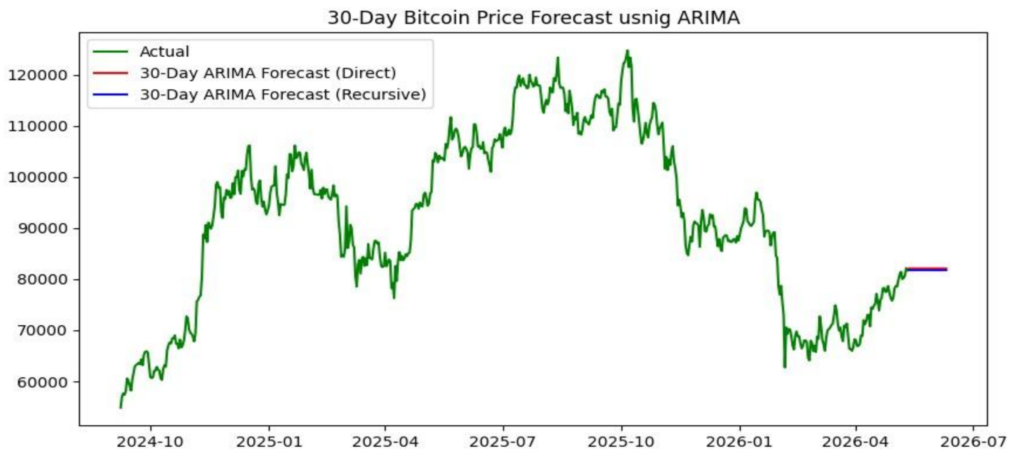

Figure 6. 30-Day Bitcoin Price Forecast using ARIMA

6. SARIMA MODEL

SARIMA (Seasonal Autoregressive Integrated Moving Average) extends the ARIMA model by incorporating seasonal components that capture periodic patterns in the data. The model is specified as SARIMA (p, d, q) (P, D, Q, s), where the additional parameters P, D, and Q represent the seasonal autoregressive, differencing, and moving average orders respectively, and s denotes the seasonal period. For this analysis, a weekly seasonal period (s=7) was selected based on the daily frequency of the data and the observation that cryptocurrency markets often exhibit weekly trading patterns driven by institutional trading schedules and weekend volatility differences.

6.1 Model Configuration

The Auto ARIMA procedure with seasonal search was used to identify the optimal SARIMA parameters. The search evaluated multiple seasonal and non-seasonal parameter combinations and determined SARIMA (1,1,0) (0,0,0) as the best configuration, with an AIC of 52,609.789. Notably, the seasonal components were found to be non-significant, resulting in a model that effectively reduces to a non-seasonal ARIMA (1,1,0). This suggests that Bitcoin daily prices do not exhibit strong weekly seasonal patterns, which is consistent with the 24/7 nature of cryptocurrency trading that differs from traditional markets with weekend closures.

6.2 Prediction Results

The SARIMA model predictions closely follow the actual Bitcoin price trajectory on the test dataset. The step-by-step forecasting approach, where the model is retrained with each new observation, enables the model to adapt to changing market conditions effectively. The prediction accuracy is comparable to the ARIMA model, which is expected given that the optimal seasonal parameters were found to be zero.

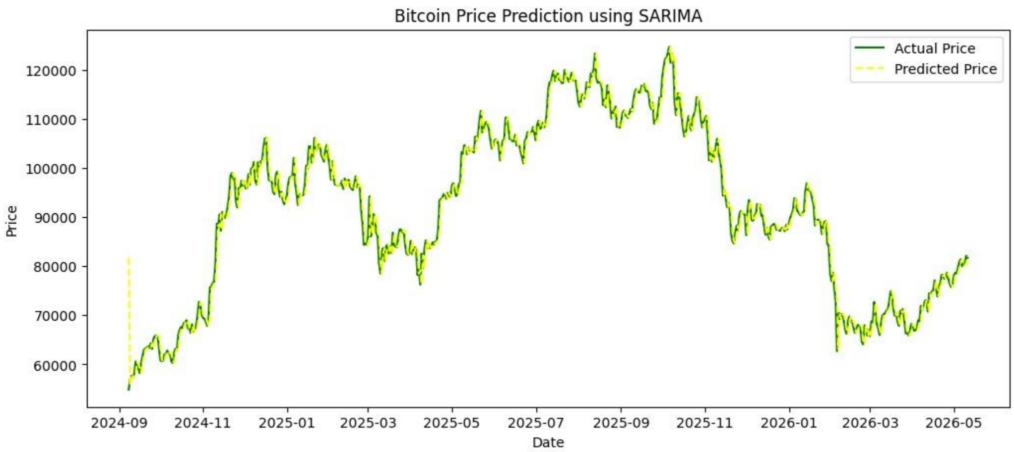

Figure 7. Bitcoin Price Prediction using SARIMA

6.3 Performance Metrics

Metric	Value	Interpretation
MSE	5,571,096	Slightly higher than ARIMA
RMSE	2,360.32	Approx. 2.6% of average BTC price
MAE	1,546.94	Similar to ARIMA performance
MAPE	1.76%	Excellent prediction accuracy
R-squared	0.981	98.1% of variance explained

Table 6. SARIMA Model Performance Metrics

6.4 30-Day Forecast

The 30-day SARIMA forecast shows a relatively flat prediction, suggesting that the model expects Bitcoin prices to remain stable in the near future. The flat nature of the forecast is characteristic of SARIMA models when applied to data without strong seasonal trends, as the model essentially captures the recent mean level without projecting significant directional movement. While this may appear conservative, it reflects the model's interpretation that the available data does not provide sufficient evidence for a strong upward or downward trend in the immediate future.

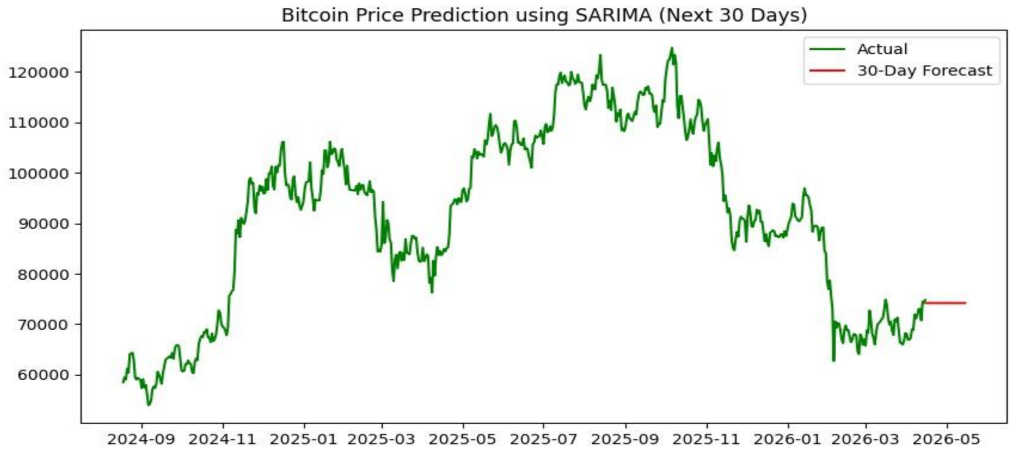

Figure 8. 30-Day Bitcoin Price Forecast using SARIMA

7. PROPHET MODEL

Prophet is a time series forecasting model developed by Facebook (Meta) that is specifically designed to handle real-world time series data with strong seasonal effects and multiple seasons of historical data. The model decomposes the time series into three core components: trend, seasonality, and holidays/events, making it particularly effective for data like cryptocurrency prices where multiple overlapping patterns exist. Prophet is robust to missing data, shifts in the trend, and outliers, and provides future predictions along with uncertainty intervals that help quantify the confidence of the forecasts.

7.1 Model Configuration

The Prophet model was configured with a changepoint prior scale of 0.5 to allow flexibility in detecting trend changes, daily seasonality enabled to capture intra-week patterns, and yearly seasonality enabled to capture annual market cycles. Additionally, trading volume was incorporated as an extra regressor to provide the model with additional information about market activity levels. The model was trained on the complete Bitcoin price dataset with volume as a supplementary feature, using the mean volume value for future predictions where volume data is not available.

7.2 Prediction Results and Decomposition

The Prophet model generates predictions along with confidence intervals that widen over time, reflecting increasing uncertainty for longer forecast horizons. The model successfully captures the overall upward trend in Bitcoin prices and provides reasonable confidence bands that encompass most of the historical price movements. The decomposition of the Prophet model reveals three key components that provide valuable insights into market structure.

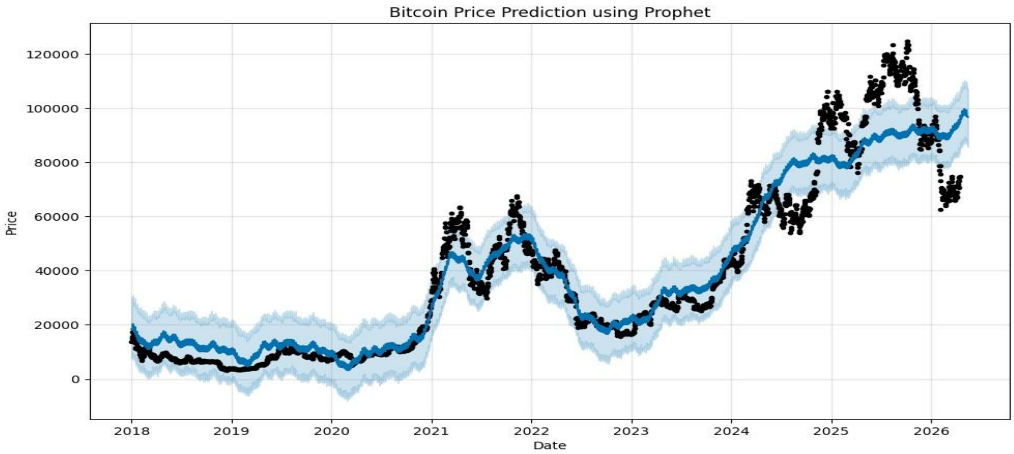

Figure 9. Bitcoin Price Prediction using Prophet

The trend component shows that the overall Bitcoin price trend is increasing over time, indicating long-term market growth despite periodic corrections. The weekly seasonality pattern suggests that Bitcoin prices generally decrease from Sunday to Monday, remain relatively stable from Monday through Friday, and then start increasing again from Friday to Saturday. The yearly seasonality analysis indicates that Bitcoin prices tend to rise during the March to May period, showing stronger market performance during these months, while periods of relative weakness tend to occur in the late summer and early fall months.

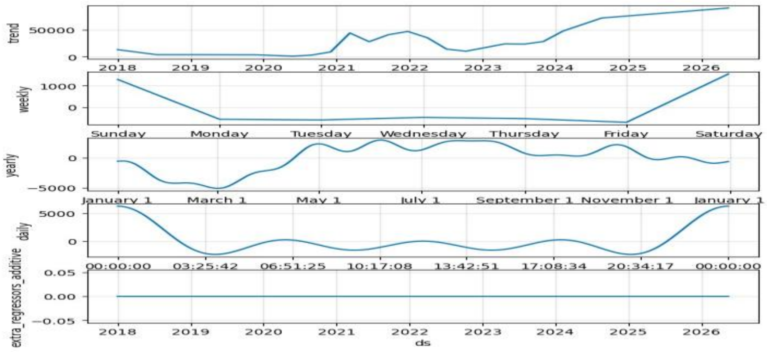

Figure 10. Prophet Model Component Decomposition

7.3 Cross-Validation Results

Cross-validation was performed using an initial training period of 730 days, with a forecast horizon of 30 days and a rolling period of 30 days between cross-validation windows. The results show that forecasting error gradually increases as the prediction horizon becomes larger, which is expected given the inherent uncertainty in cryptocurrency markets. For shorter forecasting periods of 3-7 days, the model achieved lower RMSE values around 9,480-10,550, indicating better prediction accuracy for near-term forecasts. For longer forecasting periods of 20-30 days, the RMSE values increased to approximately 13,000-14,000, reflecting the greater difficulty of predicting Bitcoin prices over extended timeframes. The MAPE values range from approximately 10.9% for 3-day forecasts to over 25% for 30-day forecasts, confirming that prediction accuracy degrades significantly with longer horizons.

Horizon	RMSE	MAE	MAPE
3 days	9,481	5,764	10.9%
7 days	10,549	6,544	13.0%
14 days	11,962	8,083	18.1%
30 days	~14,100	~10,200	~25.0%

Table 7. Prophet Cross-Validation Performance by Horizon

8. LSTM MODEL

Long Short-Term Memory (LSTM) is a specialized deep learning architecture designed for sequential data processing. Unlike traditional recurrent neural networks that struggle with long-term dependencies due to vanishing gradient problems, LSTM networks incorporate memory cells and gating mechanisms that enable them to selectively retain or discard information over extended sequences. This makes LSTM particularly well-suited for time series forecasting tasks where both short-term fluctuations and long-term trends are important for making accurate predictions. The model processes previous price sequences to capture temporal patterns in the market, and data normalization and sequence generation techniques are applied before training to improve model performance.

8.1 Model Architecture

The LSTM model was constructed with a three-layer architecture designed to capture patterns at different levels of abstraction. The first LSTM layer contains 100 units with return sequences enabled to pass the full sequence output to the next layer, followed by a 20% dropout layer to prevent overfitting. The second LSTM layer contains 60 units, also with return sequences and 20% dropout. The third LSTM layer contains 30 units with only the final output returned, followed by another dropout layer. A dense output layer with a single unit produces the price prediction. The model was compiled using the Adam optimizer with mean squared error as the loss function. Input sequences of 100-time steps were used, and the data was normalized using min-max Scaler to the range [0, 1] before training to ensure stable gradient updates.

Layer	Type	Units	Dropout
1	LSTM	100	0.2
2	LSTM	60	0.2
3	LSTM	30	0.2
4	Dense	1	-

Table 8. LSTM Model Architecture

8.2 Training Process

The model was trained for up to 50 epochs with a batch size of 50, using early stopping with a patience of 5 epochs on the validation loss to prevent overfitting. The training loss decreased rapidly in the initial epochs and stabilized after approximately 10-15 epochs, while the validation loss showed more variability, indicating the challenge of generalizing to unseen cryptocurrency data. The early stopping mechanism ensured that the model retained the best weights based on validation performance rather than continuing to train on potentially noisy patterns.

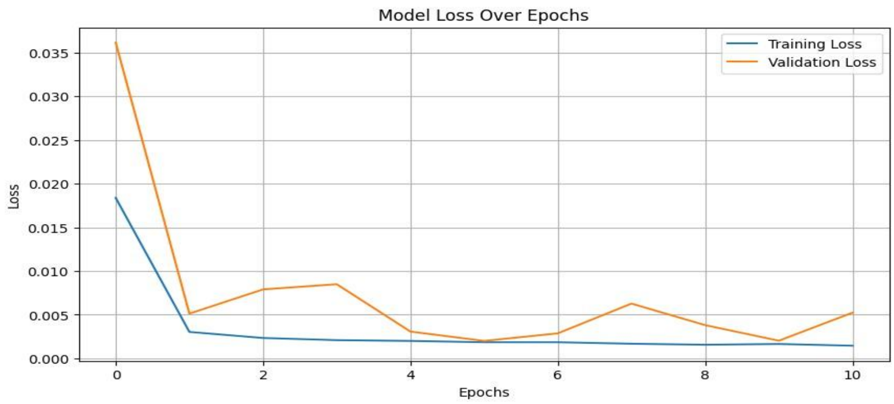

Figure 11. LSTM Training and Validation Loss Over Epochs

8.3 Prediction Results

The LSTM model was evaluated on both training and test data to assess its ability to learn patterns and generalize to new data. On the training set, the model achieved an R-squared score of 0.978 and a MAPE of 8.87%, indicating good fit to the training data. On the test set, the model achieved an R-squared score of 0.900 and a MAPE of 4.64%. The higher error metrics on the test set compared to the training set reflect the inherent challenge of extrapolating to unseen market conditions, particularly in the highly volatile cryptocurrency market. The visual comparison of actual versus predicted prices on the test data shows that the LSTM model captures the general trend direction but tends to smooth out extreme price movements.

Metric	Training Set	Test Set
MSE	8,161,623	29,660,018
RMSE	2,856.86	5,446.10
MAE	1,946.61	4,200.93
MAPE	8.87%	4.64%
R-squared	0.978	0.900

Table 9. LSTM Model Performance Metrics

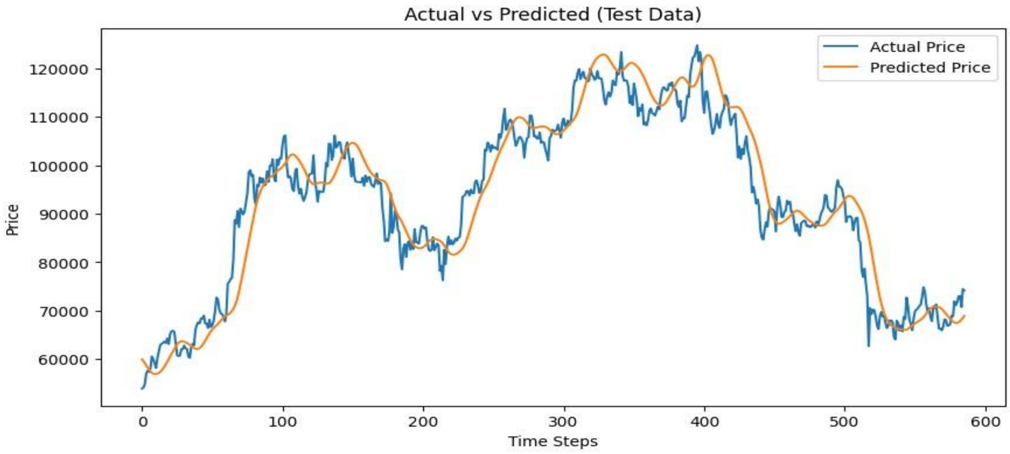

Figure 12. Actual vs Predicted Bitcoin Prices using LSTM (Test Data)

8.4 30-Day Forecast

The LSTM 30-day forecast was generated using a recursive prediction approach, where the model predicts one day at a time and appends the predicted value to the input sequence for the next prediction. The forecast shows a relatively stable trend around the recent market price range, with the model predicting a slight downward movement after a small initial increase. This suggests that the LSTM model expects Bitcoin prices to remain within a narrow range in the near future, without projecting major directional moves. The conservative nature of the forecast reflects the model's learned representation of recent market conditions and its tendency toward mean-reverting predictions when faced with uncertainty.

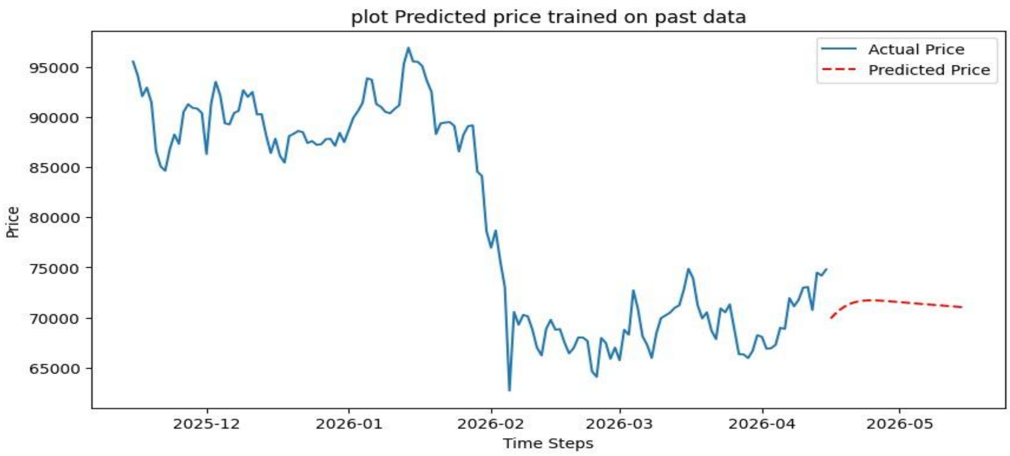

Figure 13. 30-Day Bitcoin Price Forecast using LSTM

9. MODEL COMPARISON

A comprehensive comparison of all four models reveals distinct strengths and weaknesses for each approach. The ARIMA and SARIMA models achieved the highest R-squared scores (0.985 and 0.981 respectively) and the lowest MAPE values (1.68% and 1.76%), making them the most accurate models for short-term Bitcoin price prediction on the test dataset. The LSTM model achieved a test R-squared of 0.900 with a MAPE of 4.64%, showing respectable performance but falling short of the statistical models in terms of raw prediction accuracy. The Prophet model, evaluated through cross-validation rather than a simple train-test split, showed increasing error with longer forecast horizons, with MAPE values ranging from approximately 10.9% for 3-day forecasts to over 25% for 30-day forecasts.

Metric	ARIMA	SARIMA	LSTM (Test)
RMSE	2,097	2,360	5,446
MAE	1,507	1,547	4,201
MAPE	1.68%	1.76%	4.64%
R-squared	0.985	0.981	0.900

Table 10. Model Performance Comparison

It is important to note that the superior performance of ARIMA and SARIMA is partly attributable to the rolling forecast approach used for these models, where each prediction incorporates the most recent actual value. The LSTM model, in contrast, was evaluated using a static test set without rolling updates, which places it at a disadvantage in terms of short-term accuracy. However, the LSTM model offers potential advantages in capturing complex non-linear patterns that statistical models may miss, particularly during regime changes or periods of unusual market behavior. The Prophet model provides unique value through its decomposition capabilities, offering interpretable insights into trend, seasonality, and the uncertainty of predictions that the other models do not readily provide.

10. CONCLUSION

This project has systematically applied and compared four distinct modeling approaches for Bitcoin price prediction: ARIMA, SARIMA, Prophet, and LSTM. The analysis was conducted on over eight years of daily Bitcoin price data, providing a comprehensive foundation for model training and evaluation across multiple market cycles. The exploratory data analysis revealed that Bitcoin prices are characterized by high volatility, non-stationarity, and the presence of both short-term fluctuations and long-term trends, with key technical indicators such as moving average crossovers providing valuable signals about market direction.

Among the models evaluated, ARIMA and SARIMA demonstrated the strongest performance for short-term forecasting, achieving R-squared values above 0.98 and MAPE values below 2%. These statistical models are particularly effective when used with a rolling forecast approach that continuously incorporates the latest market data. The Prophet model added value through its ability to decompose the time series into interpretable components, revealing both weekly and yearly seasonal patterns in Bitcoin prices, while providing uncertainty estimates that are crucial for risk management. The LSTM model showed competitive performance with a test R-squared of 0.900 and demonstrated the potential to capture complex non-linear patterns in the data, though its accuracy was limited by the static evaluation approach and the inherent challenges of training deep learning models on financial time series.

ARIMA or SARIMA is recommended for short-term day-ahead forecasting where maximum accuracy is needed. Prophet is preferred when interpretability and uncertainty quantification are important, particularly for longer forecast horizons. LSTM may be most valuable when the goal is to capture complex, non-linear market dynamics, especially during periods of unusual market behavior where traditional statistical models may underperform. Future work could explore ensemble methods that combine the strengths of multiple models, as well as the incorporation of additional features such as onchain metrics, sentiment analysis data, and macroeconomic indicators to further improve prediction accuracy.